

1) write a python program to declare variables , perform arithmetic operations and display results ?

=====

```
a = 20
```

```
b = 10
```

```
# addition
```

```
print(a+b)
```

```
print(a-b)
```

```
print(a*b)
```

```
print(a/b)
```

```
# area of rectangle
```

```
length = 10
```

```
breadth = 20
```

```
area = length * breadth
```

```
print(area)
```

.....:OUTPUT:.....

```
30
```

```
10
```

```
200
```

```
2.0
```

```
200
```

.....

```
=====
#2) Create a program to check if a number is even or odd using if-else ?
=====
```

```
num = int(input("Enter a number: "))
```

```
if num % 2 == 0:
```

```
    print("its a even number")
```

```
else:
```

```
    print("its a odd number")
```

```
.....:OUTPUT:.....
```

```
Enter a number: 10
```

```
its a even number
```

```
.....
```

```
=====
#3) write a python program to print the first n fibonacci numbers using a for loop ?
=====
```

```
def fibonacci(num):
```

```
    a = 0
```

```
    b = 1
```

```
    for i in range(num):
```

```
        yield a
```

```
        a, b = b, a+b
```

```
print(list(fibonacci(10)))
```

```
.....:OUTPUT:.....
```

```
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

```
.....
```

```
=====
# 4) Implement a program that accepts a String and counts the number of Vowels and
Consonants ?
=====
```

```
string = "welcome to Python Session"
```

```
vowel_count = [ ]
```

```
consonant_count = [ ]
```

```
for i in range(0,len(string),1):
```

```
    if string[i] in "aeiouAEIOU":
```

```
        vowel_count.append(string[i])
```

```
    else:
```

```
        consonant_count.append(string[i])
```

```
print("The Vowel count in a string is :",len(vowel_count))
```

```
print("The Consonant count in a string is :",len(consonant_count))
```

```
:::::::::::::::::::::::::::::OUTPUT:::::::::::::::::::::::::::::::::::::
```

```
The Vowel count in a string is : 8
```

```
The Consonant count in a string is : 17
```

```
::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::
```

```
=====
#5) Create a Program to store student details in a dictionary and retrieve details based
on user input ?
=====
```

```
student = {"name":"Jeevan", "age":20 , "class":"BCA"}
```

```
# adding details and Updating
```

```
student.update({"class":"MCA", "blood":"O+ve"})
```

```
print(student)
```

```
# Retrieve details based on input
```

```
print(student.get("class"))
```

```
.....OUTPUT:.....
```

```
{'name': 'Jeevan', 'age': 20, 'class': 'MCA', 'blood': '0+ve'}
```

```
MCA
```

```
.....
```

```
=====
```

```
#6) Demonstrate the use of Break , continue and Pass Statements in Loops ?
```

```
=====
```

```
for i in range(1,10):
```

```
    if i == 2 :
```

```
        continue      #continue >> it stops the current iteration >> skips value 2
```

```
    if i == 5 :
```

```
        pass          #pass > No effect
```

```
    print("i = " , i)
```

```
    if i == 8:
```

```
        break          #break >> it stops the Complete Iteration >> exits the loop
```

```
.....OUTPUT:.....
```

```
i = 1
```

```
i = 3
```

```
i = 4
```

```
i = 5
```

```
i = 6
```

```
i = 7
```

```
i = 8
```

```
.....
```

```
=====
# 7) write a Program to Create Numpy arrays , perform element-wise Operations, and
Reshape arrays ?
=====
```

```
# creating numpy arrays
```

```
import numpy as np
```

```
a = np.array([1,3,5,7])
```

```
b = np.array([2,4,6,8])
```

```
print("Created Numpy Array")
```

```
print(a)
```

```
print(b)
```

```
# perform element wise operations
```

```
print("Element wise Operation")
```

```
print(np.add(a,b))          # or print(a+b)
```

```
print(np.subtract(a,b))
```

```
# Reshape arrays
```

```
print("Reshaping Array with 10 elements")
```

```
new = np.arange(10)
```

```
new = new.reshape(2,5)
```

```
print(new)
```

```
:::::::::::::::::::::::::::::::OUTPUT:::::::::::::::::::::::::::::::::::::
```

```
Created Numpy Array
```

```
[1 3 5 7]
```

```
[2 4 6 8]
```

```
Element wise Operation
```

```
[ 3  7 11 15]
```

```
[-1 -1 -1 -1]
```

Reshaping Array with 10 elements

```
[[0 1 2 3 4]
```

```
[5 6 7 8 9]]
```

```
.....
```

```
=====
```

8) Create a PANDAS Series and Perform Indexing , Slicing and Querying Operations ?

```
=====
```

Create Pandas series

```
import pandas as pd
```

```
new = pd.Series( [1,2,3,4,5],
```

```
                index=["a","b","c","d","e"] )
```

```
print("Output after Creating Pandas series : ")
```

```
print(new)
```

indexing

```
print("Indexing Operation : ")
```

```
print(new["a"])
```

```
print(new["e"])
```

slicing

```
print("Slicing Operation : ")
```

```
print(new["b":"d"])
```

Query

```
print("Query Operation : ")
```

```
print(new["a"])
```

```
print(new[new < 3])
```

.....OUTPUT:.....

Output after Creating Pandas series :

a 1

b 2

c 3

d 4

e 5

dtype: int64

Indexing Operation :

1

5

Slicing Operation :

b 2

c 3

d 4

dtype: int64

Query Operation :

1

a 1

b 2

dtype: int64

.....

```
=====

## 9) Load a Dataset into Pandas DataFrame and Perform Sorting and Filtering
Operations ?

=====
```

```
import pandas as pd

# create sample CSV

df = pd.DataFrame({
    "name":["Ajay","Bhavani","Chetan","Divya"],
    "score":[88, 55, 78, 95],
    "age":[20,22,21,23]
})

df.to_csv("sample_data.csv", index=False)

df = pd.read_csv("sample_data.csv")

print("Sorted by score:\n", df.sort_values("score", ascending=False))

print("Filter score >= 80:\n", df[df["score"] >= 80])
```

```
:::::::::::::::::::::::::::::OUTPUT:::::::::::::::::::::::::::::::::::::
```

Sorted by score:

	name	score	age
3	Divya	95	23
0	Ajay	88	20
2	Chetan	78	21
1	Bhavani	55	22

Filter score >= 80:

	name	score	age
0	Ajay	88	20
3	Divya	95	23

.....

=====

#10) Write a program to handle missing values by filling them with mean/median values?

=====

```
import numpy as np
```

```
import pandas as pd
```

```
df = pd.DataFrame({"Even":[2,4,np.nan,8,10],
```

```
                  "Odd":[1,3,np.nan,7,9]})
```

```
print("Original values\n",df)
```

```
df_mean = df.fillna(df.mean())
```

```
df_median = df.fillna(df.median())
```

```
print("Mean filled values\n",df_mean)
```

```
print("Median filled values\n",df_median)
```

.....:OUTPUT:.....

Original values

	Even	Odd
--	------	-----

0	2.0	1.0
---	-----	-----

1	4.0	3.0
---	-----	-----

2	NaN	NaN
---	-----	-----

3	8.0	7.0
---	-----	-----

4	10.0	9.0
---	------	-----

Mean filled values

Even Odd

0 2.0 1.0

1 4.0 3.0

2 6.0 5.0

3 8.0 7.0

4 10.0 9.0

Median filled values

Even Odd

0 2.0 1.0

1 4.0 3.0

2 6.0 5.0

3 8.0 7.0

4 10.0 9.0

.....

=====

=====

PART-B

=====

11) String Operations : Write a Program to Count the occurrences of each word in a given String ?

=====

str = input("Enter a string : ")

f = { }

```
for i in str:
```

```
    if i in f:
```

```
        f[i] = f[i] + 1
```

```
    else:
```

```
        f[i] = 1
```

```
print(f)
```

```
:::::::::::::::::::::::::::::OUTPUT:::::::::::::::::::::::::::::::::::::
```

```
Enter a string : WELCOME
```

```
{'W': 1, 'E': 2, 'L': 1, 'C': 1, 'O': 1, 'M': 1}
```

```
::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::::
```

```
=====
```

```
# 12) Implement a Program to Insert , delete and Update elements in a list ?
```

```
=====
```

```
list = [10,20,30,50]
```

```
print("Original list : ",list)
```

```
# insert
```

```
list.insert(3,40)
```

```
print("after insert operation :",list)
```

```
# delete
```

```
list.remove(30)
```

```
print("after delete operation : ",list)
```

```
# update
```

```
list[1]= 100
```

```
print("after update operation :",list)
```

```
.....:OUTPUT:.....
```

```
Original list : [10, 20, 30, 50]
```

```
after insert operation : [10, 20, 30, 40, 50]
```

```
after delete operation : [10, 20, 40, 50]
```

```
after update operation : [10, 100, 40, 50]
```

```
.....
```

```
=====
```

```
# 13) Create a Dictionary with Employee details and perform CRUD operations ?
```

```
=====
```

```
Employee = { }
```

```
# Create
```

```
Employee = {"Name": "Jeevan", "Salary": 10000, "Age": 25}
```

```
print(Employee)
```

```
# Read
```

```
print(Employee.get("Name"))
```

```
#Update
```

```
Employee["Name"] = "Vikrant"
```

```
print(Employee)
```

```
# Delete
```

```
del Employee["Age"]
```

```
print(Employee)
```

```
.....:OUTPUT:.....
```

```
{'Name': 'Jeevan', 'Salary': 10000, 'Age': 25}
```

Jeevan

```
{'Name': 'Vikrant', 'Salary': 10000, 'Age': 25}
```

```
{'Name': 'Vikrant', 'Salary': 10000}
```

.....

=====

14) Implement a Program to generate Prime Numbers upto n using a Generator function ?

=====

```
# prime generator
```

```
def generate_primes(n):
```

```
    for num in range(2, n + 1):
```

```
        for i in range(2, int(num ** 0.5) + 1):
```

```
            if num % i == 0:
```

```
                break
```

```
            else:
```

```
                yield num # yield returns one prime at a time
```

```
# Example usage
```

```
range = int(input("Enter the range: "))
```

```
print(f"Prime numbers up to {range}:")
```

```
for prime in generate_primes(range):
```

```
    print(prime, end=" ")
```

.....:OUTPUT:.....

Enter the range: 10

Prime numbers up to 10:

2 3 5 7

.....

```
=====
# 15) Write a Python program to Calculate the Factorial of a Number using Recursion ?
=====
```

```
def fact(n):
    if n <= 1:
        return 1
    return n * fact(n-1)
```

```
print("5! =", fact(5))
```

```
=====
Output :: 5! = 120
```

```
##### OR #####
```

```
=====
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n - 1)
```

```
# Take input from user
```

```
num = int(input("Enter a number: "))
```

```
# Check for negative input
```

```
if num < 0:
```

```
    print("Factorial does not exist for negative numbers.")
```

```
else:
```

```
    print(f"The factorial of {num} is {factorial(num)}")
```

.....:OUTPUT:.....

Enter a number: 5

The factorial of 5 is 120

.....

=====

16) write a Program to Read and write students marks into a text file and display the contents ?

=====

Read and Write

students = [("Jeevan", 88),("Vikrath",95),("Soumya",60)]

with open("students.txt","w") as file:

for name,marks in students:

file.write(name+"," +str(marks)+"\n")

read and Display

with open("students.txt","r") as file:

for students in file:

print(students)

.....:OUTPUT:.....

Jeevan,88

Vikrath,95

Soumya,60

.....

=====

17) Plot a line graph and a Bar Graph using Matplotlib ?

=====

```
import matplotlib.pyplot as plt
```

```
# line plot
```

```
x = [1,2,3,4]
```

```
y = [10,20,25,30]
```

```
plt.figure()
```

```
plt.plot(x, y)
```

```
plt.title("Line Plot")
```

```
plt.xlabel("X")
```

```
plt.ylabel("Y")
```

```
plt.show()
```

```
# bar chart
```

```
plt.figure()
```

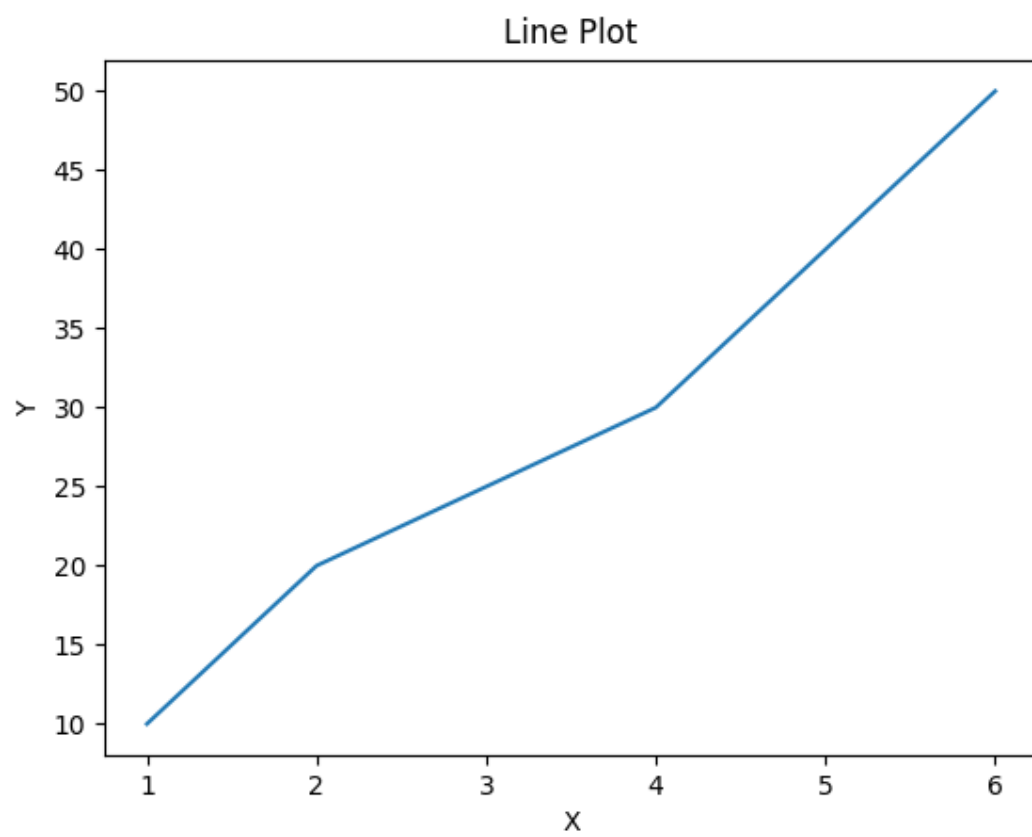
```
plt.bar(["Maths","Science","English"], [45,80,95])
```

```
plt.title("Bar Chart of Subject and Marks")
```

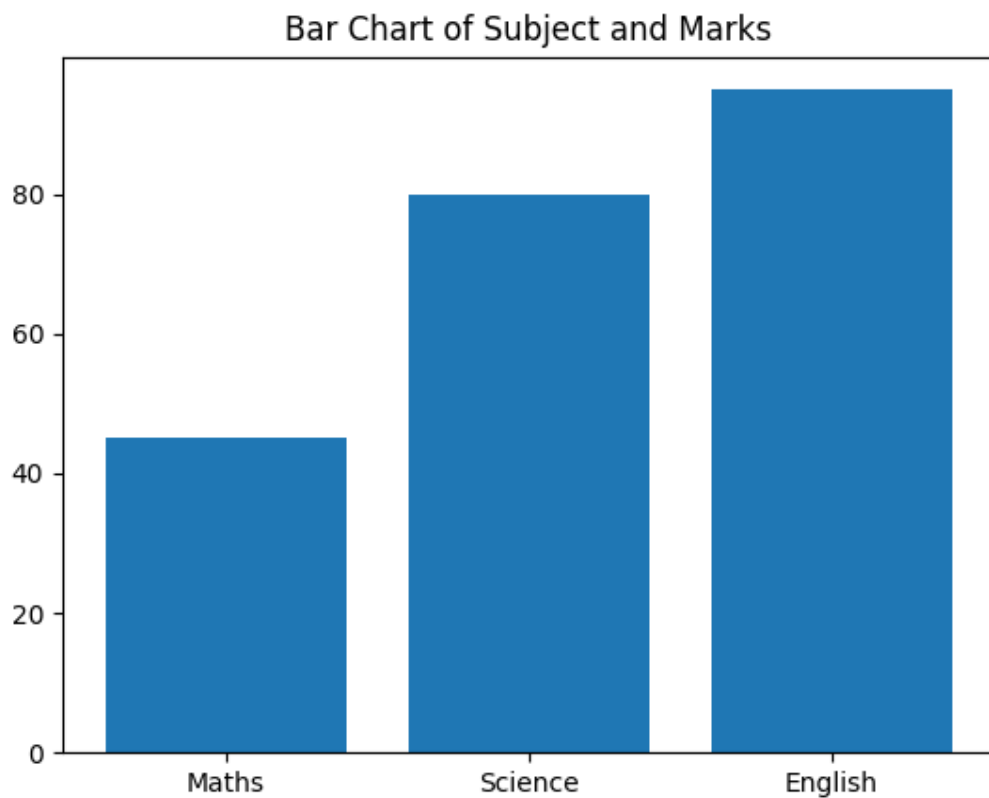
```
plt.show()
```

```
.....:OUTPUT:.....
```

Line plot Graph >> draw or take Printout and Paste it in Record Book



Bar Graph >> draw or take Printout and Paste it in Record Book



.....

=====

18) write a Program to load a CSV file into Pandas and perform basic Data Analysis ?

=====

load CSV and basic analysis

import pandas as pd

use sample_data.csv created earlier

df = pd.read_csv("sample_data.csv")

print("Head:\n", df.head())

```
print("Describe:\n", df.describe())

print("Value counts for name:\n", df["name"].value_counts())
```

.....OUTPUT:.....

Head:

	name	score	age
0	Ajay	88	20
1	Bhavani	55	22
2	Chetan	78	21
3	Divya	95	23

Describe:

	score	age
count	4.000000	4.000000
mean	79.000000	21.500000
std	17.454703	1.290994
min	55.000000	20.000000
25%	72.250000	20.750000
50%	83.000000	21.500000
75%	89.750000	22.250000
max	95.000000	23.000000

Value counts for name:

Name	
Ajay	1
Bhavani	1
Chetan	1
Divya	1

Name: count, dtype: int64

.....

=====

19) Implement a Program to group a dataset by Category and calculate Summary statistics ?

=====

groupby summary

import pandas as pd

```
df = pd.DataFrame({  
    "category":["X","Y","X","Y","X"],  
    "value":[10,20,30,40,50]  
})
```

```
grouped = df.groupby("category")["value"].agg(["count","mean","sum"])
```

```
print(grouped)
```

.....OUTPUT:.....

	Count	mean	sum
category			
X	3	30.0	90
Y	2	30.0	60

.....

=====

#20) Load a Dataset from Scikit-learn and Display its basic properties ?

=====

from sklearn import datasets

```
iris = datasets.load_iris()
```

```
X = iris.data
y = iris.target
```

```
print("Shape:", X.shape)
```

```
print("Feature names:", iris.feature_names)
```

```
print("Target names:", iris.target_names)
```

```
# show first 3 rows
```

```
print("First 3 rows:\n", X[:3])
```

```
.....:OUTPUT:.....
```

```
Shape: (150, 4)
```

```
Feature names: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
```

```
Target names: ['setosa' 'versicolor' 'virginica']
```

```
First 3 rows:
```

```
[[5.1 3.5 1.4 0.2]
```

```
[4.9 3. 1.4 0.2]
```

```
[4.7 3.2 1.3 0.2]]
```

```
.....
```

```
=====
```